Exact Orders, Shift Optimality, and Zero Asymptotics for q-Exponential Hankel Determinants

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Abstract

We study shifted Hankel determinants of the remainder sequence $v_0 = \mu - 1$, $v_k = (q^k - 1)v_{k-1} - \alpha^k$ for the q-exponential. A two-node formal moment expansion gives the exact generic order at $q = 0$, explicit lowest coefficients, and the polynomial degree. At $\mu = E_q(\alpha)$, positive geometric moments give sharp size estimates. A degree budget for a prescribed cyclotomic divisor proves that proportional index shifts cannot improve the critical values 12/7 and 6 in the associated cubic norm comparison; these arithmetic values are already known. For real $q > 1$, we then extend the moment representation to complex α, remove the exact forced zeros, and obtain a Schur expansion of the normalized entire determinant. After scaling by q^{n+r}, this function converges locally uniformly, uniformly in all shifts $r \geq 0$, to $\prod_{\ell \geq 1}(1 + zq^{-\ell})^{\ell}$, with an explicit geometric error. This gives a full leading asymptotic and its first correction at fixed α, and determines the limiting zero clusters. Using classical q-Meixner polynomials, we reduce a special shifted determinant from size n to size r and locate the first additional zero to relative error $O(q^{-n})$ for every fixed shift. We also resolve the second unshifted cluster into two real simple zeros with distinct superexponential displacement scales. These analytic refinements do not assert a lower general irrationality threshold. Modular specializations are recorded separately as consequences of existing transcendence theory.

Keywords. q-exponential; Hankel determinant; cyclotomic divisibility; Schur functions; q-Meixner polynomials; zero asymptotics.

1 Introduction and principal results

For $|q| > 1$, consider the entire function

$$E_q(z) = F_q(z; 1) = \sum_{k=0}^{\infty} \frac{z^k}{\prod_{j=1}^{k}(q^j - 1)}. \tag{1.1}$$

Krattenthaler, Rochev, Väänänen, and Zudilin [1] use Hankel determinants of remainder sequences to study the arithmetic of this function and related q-series. Their construction separates three cubic growth quantities: smallness at the distinguished value, degree of a common polynomial divisor, and degree or conjugate growth of the determinant. For the remainder sequence used here, these quantities are especially amenable to an exact analysis of an index shift.

Throughout the algebraic sections, q, α, μ are independent indeterminates, and

$$R = \mathbb{Z}[q, \alpha, \mu], \qquad v_0 = \mu - 1, \qquad v_k = (q^k - 1)v_{k-1} - \alpha^k \quad (k \geq 1). \tag{1.2}$$

For integers $n \geq 1$ and $r \geq 0$, put

$$V_n^{(r)} = \det(v_{i+j+r})_{0 \leq i,j < n}, \qquad V_n = V_n^{(0)}. \tag{1.3}$$

At the specialization $\mu = E_q(\alpha)$, direct solution of (1.2) gives

$$v_k = \sum_{\ell=k+1}^{\infty} \frac{\alpha^\ell}{\prod_{j=k+1}^{\ell}(q^j - 1)}. \tag{1.4}$$

We use (1.4) analytically only where its parameters are specified. In particular, the positivity and exact smallness results below assume $q > 1$ and $\alpha > 0$.

The first part of this article gives complete proofs of the exact algebraic endpoints, the prescribed cyclotomic divisor, and the restriction on the associated shift comparison. The second part studies the determinant as an entire function of α and resolves zero displacements that the cubic arithmetic estimates cannot detect. The two parts address different scales: the arithmetic comparison uses terms of order n^3, whereas the analytic refinements determine constants, geometric corrections, and exponentially small separations of zeros.

The thresholds 12/7 and 6 are not new arithmetic bounds. Rochev [2, Corollary 1.1] already obtains the corresponding 7/12 and 1/6 conditions in his height notation. Our optimality statement is restricted to the divisor and norm comparison specified below. The analytic arguments use classical Schur identities [3] and q-Meixner orthogonality [4]; these inputs are explicitly distinguished from the determinant identities and zero estimates proved here.

Theorem 1.1 (Exact algebraic endpoints)**.** *For all $n \geq 1$ and $r \geq 0$, in $\mathbb{Q}(\alpha, \mu)[q]$ one has*

$$e_{n,r} := \operatorname{ord}_q V_n^{(r)} = \binom{n}{3} + r \left\lfloor \frac{(n-1)^2}{4} \right\rfloor, \tag{1.5}$$

$$D_{n,r} := \deg_q V_n^{(r)} = \sum_{i=0}^{n-1} \frac{(2i+r)(2i+r+1)}{2}. \tag{1.6}$$

The lowest coefficients are given explicitly in Proposition 2.2. Consequently, $q^{e_{n,r}}$ divides $V_n^{(r)}$ in R. The equality in (1.5) *is generic: after specializing α or μ, the order may increase.*

Theorem 1.2 (Positive moments and exact size)**.** *Let $q > 1$, $\alpha > 0$, and $\mu = E_q(\alpha)$. Define*

$$x_j = \alpha q^{-j}, \qquad w_j = \alpha q^{-j} \prod_{\ell=1}^{j-1}(1 + \alpha q^{-\ell}) \quad (j \geq 1). \tag{1.7}$$

Then $v_k = \sum_{j \geq 1} w_j x_j^k$ for every $k \geq 0$, and $V_n^{(r)} > 0$ for every $n \geq 1, r \geq 0$. With

$$K_{n,r} = \frac{n(n+1)(2n+1)}{6} + \frac{rn(n+1)}{2}, \qquad Z_{n,r} = \alpha^{n^2+rn} q^{-K_{n,r}}, \tag{1.8}$$

we have

$$Z_{n,r} \prod_{k=1}^{n-1}(1 - q^{-k})^{2(n-k)} \leq V_n^{(r)} \leq Z_{n,r} \mu^n \prod_{m=1}^{n}(1 - q^{-m(m+r)})^{-1}. \tag{1.9}$$

In particular, uniformly in integers $r \geq 0$,

$$\log V_n^{(r)} = -K_{n,r} \log q + (n^2 + rn) \log \alpha + O_{q,\alpha}(n). \tag{1.10}$$

To state the third result, we specify the divisor rather than invoking an unspecified "full content." Write

$$S_d(n) = \sum_{i=0}^{n-1} \left\lfloor \frac{i}{2d} \right\rfloor, \tag{1.11}$$

$$T_d(n,p) = \sum_{i=0}^{n-1} \min\left\{ \left\lfloor \frac{i}{d} \right\rfloor, \left\lfloor \frac{i+p}{2d} \right\rfloor \right\}, \tag{1.12}$$

$$\nu_d(n,r) = \max_{p+p'=r} (T_d(n,p) + T_d(n,p')), \tag{1.13}$$

$$\mathcal{D}_{n,r}(q) = q^{e_{n,r}} \prod_{d\geq 1} \Phi_d(q)^{\nu_d(n,r)}, \tag{1.14}$$

where p, p' are nonnegative integers and Φ_d is the d-th cyclotomic polynomial. The product is finite because $\nu_d(n,r) = 0$ for $d \geq n$.

Theorem 1.3 (Optimality in the prescribed shift comparison)**.** *The polynomial $\mathcal{D}_{n,r}$ divides $V_n^{(r)}$ in R. Fix $t \geq 0$ and $r = \lfloor tn \rfloor$. The exact smallness and degree constants are*

$$A(t) = \frac{1}{3} + \frac{t}{2}, \qquad C(t) = \frac{2}{3} + t + \frac{t^2}{2}. \tag{1.15}$$

Every accumulation point B of $n^{-3} \deg_q \mathcal{D}_{n,\lfloor tn \rfloor}$ satisfies

$$B \leq \overline{B}(t) := \frac{1}{4} + \frac{3t}{8} + \frac{t^2}{8}. \tag{1.16}$$

For such a B, the rationality-exclusion critical value of the standard cubic comparison satisfies

$$\tau_1(t;B) := \frac{A(t)+C(t)}{A(t)+B} \geq \frac{24+36t+12t^2}{14+21t+3t^2} \geq \frac{12}{7}. \tag{1.17}$$

The last inequality is strict for $t > 0$. For quadratic-field exclusion,

$$\tau_2(t;B) := \frac{A(t)+C(t)}{A(t)-C(t)+2B} \geq 6 \tag{1.18}$$

whenever its denominator is positive. At $t = 0$, $B = 1/4$, and both unshifted critical values are attained.

The term *critical value* in this theorem refers to a particular sufficient-condition calculation, defined by (1.17) and (1.18). It is not a boundary between rational and irrational values of $E_q(\alpha)$. Extra arithmetic divisors, specializations, alternative determinants, and stronger conjugate estimates are outside this comparison. Using accumulation points in the statement avoids assuming the existence of a shifted divisor-degree limit that is not needed for the argument.

1.1 Analytic refinements and their scope

For the remainder of this introduction let $q > 1$, $p = q^{-1}$, and write $H_{n,r}(\alpha) = V_n^{(r)}(q, \alpha, E_q(\alpha))$. After removal of its forced factors

$$\mathcal{L}_{n,r}(\alpha) = \alpha^{n^2+rn} \prod_{m=1}^{n-1} (1+\alpha p^m)^{n-m},$$

we normalize by a positive geometric Cauchy determinant $\mathcal{C}_{n,r}$; see (7.7). The quotient $\Psi_{n,r} = H_{n,r}/(\mathcal{L}_{n,r}\mathcal{C}_{n,r})$ is entire and equals one at the origin. The following are the principal analytic conclusions of this revision.

1. **A uniform entire scaling limit.** For every $R > 0$,

$$\sup_{r\geq 0,\,|z|\leq R} \left|\Psi_{n,r}(p^{-n-r}z) - \Phi_p(z)\right| = O_{p,R}(p^n), \qquad \Phi_p(z) = \prod_{\ell\geq 1}(1+zp^\ell)^\ell.$$

   Thus the kth limiting cluster has exactly k zeros counted with multiplicity. At fixed α, the same estimate yields an explicit leading product and its first correction (theorems 8.3 and 8.6).

2. **The first additional zero for every fixed shift.** Let $\pi_r = \prod_{j=1}^{r}(1-p^j)$ and $P_\infty = \prod_{j\geq 1}(1-p^j)$. For each fixed $r \geq 0$, the nearest zero of $\Psi_{n,r}$ is real, simple, and satisfies

$$\beta_{n,r} = -q^{n+r+1} - \frac{p^{n^2-n-r-1}}{P_\infty \pi_r}\left(1 + O_{p,r}(p^n)\right).$$

   An exact $r \times r$ determinant identity supplies the displacement constant (theorems 10.3 and 10.5).

3. **Asymmetric splitting of the second unshifted cluster.** For all sufficiently large n, the two zeros near $-q^{n+2}$ are real and simple. Writing them as $-q^{n+2}-\delta_n^-$ and $-q^{n+2}+\delta_n^+$, one has

$$\delta_n^- \sim \frac{p^{n^2-3n-1}}{(1-p)^2 P_\infty}, \qquad \delta_n^+ \sim \frac{(1-p)p^{n^2+n-2}}{P_\infty}.$$

   Their separation scales follow from an exact product at $\mu = 0$, a trace identity, and classical q-Meixner polynomials (theorem 9.5).

All asymptotic statements hold for fixed $q > 1$. Uniformity in r is asserted for the scaling limit, whereas the precise displacement of the first zero holds for each fixed r. We do not infer reality or individual splitting laws for all higher clusters from their limiting multiplicities. These conclusions refine the analytic structure of the same determinants; they do not change the cubic arithmetic exponents or prove a new general irrationality criterion.

2 Formal moments, exact orders, and extreme coefficients

2.1 A two-node formal moment expansion

Set $K = \mathbb{Q}(\alpha, \mu)$ and work in the q-adically complete ring $K[[q]]$. For $j \geq 0$, define

$$u_j = \frac{q^j}{\prod_{\ell=1}^{j}(1-q^\ell)}, \qquad b_j = -\frac{\alpha q^j}{\prod_{\ell=0}^{j}(1+\alpha q^\ell)}. \tag{2.1}$$

The sums $U = \sum_{j\geq 0} u_j$ and $\sum_{j\geq 0} b_j$ converge formally. Moreover, $U(0) = 1$, so U is invertible. Put

$$a(q) = \frac{\mu - 1 - \sum_{j\geq 0} b_j}{U}, \qquad a_j = a(q)u_j. \tag{2.2}$$

Then

$$a(0) = \mu - \frac{1}{1+\alpha}, \qquad \operatorname{ord}_q a_j = \operatorname{ord}_q b_j = j. \tag{2.3}$$

Here $1+\alpha$ is nonzero in K; the representation is used generically, not as a direct formula at $\alpha = -1$.

Lemma 2.1. *For every $k \geq 0$,*

$$v_k = \sum_{j\geq 0}\left[a_j(-q^j)^k + b_j(\alpha q^j)^k\right] \quad \text{in } K[[q]]. \tag{2.4}$$

Proof. Each summand has order at least $j(k+1)$, so the expression is well defined. Its value at $k=0$ is $\mu-1$ by (2.2). The coefficient identities

$$\begin{aligned} a_j(1-q^j) &= qa_{j-1} && (j\geq 1),\\ b_j(1+\alpha q^j) &= qb_{j-1} && (j\geq 1),\\ b_0(1+\alpha) &= -\alpha \end{aligned}$$

show, on shifting the summation index, that the expression on the right of (2.4) satisfies $v_k+v_{k-1}=q^kv_{k-1}-\alpha^k$. For the first family, the $j=0$ contribution to v_k+v_{k-1} vanishes; for the second family it is $-\alpha^k$. Thus the initial value and recurrence agree with (1.2), proving the identity. □

There are two nodes at each level j:

$$x_{j,-}=-q^j, \qquad x_{j,+}=\alpha q^j, \tag{2.5}$$

with weights a_j and b_j. If two distinct nodes have levels $j\leq k$, then their difference has order j. At the same level this follows from $x_{j,+}-x_{j,-}=(1+\alpha)q^j$; at different levels it follows from the nonzero coefficient of the smaller power of q.

2.2 Minimal configurations and the generic order

Proof of (1.5). Apply the finite Cauchy–Binet identity to truncations of (2.4), and then pass coefficientwise to the formal limit. For a subset of n nodes, its contribution is

$$\prod_{i=1}^{n} w_i x_i^r \prod_{i<j}(x_j-x_i)^2. \tag{2.6}$$

Write the selected levels in increasing order as $\ell_1\leq\cdots\leq\ell_n$. A level occurs at most twice, and the order of (2.6) is exactly

$$\sum_{i=1}^{n}[r+2(n-i)+1]\ell_i. \tag{2.7}$$

All coefficients in this level cost are positive. Every allowable list satisfies $\ell_i\geq\lfloor(i-1)/2\rfloor$, and equality in each position is attainable. This is therefore the unique minimizing list of levels. The formal limit is legitimate: only finitely many node subsets can have order below a prescribed bound.

The elementary sums

$$\begin{aligned} \sum_{i=1}^{n}\left\lfloor\frac{i-1}{2}\right\rfloor &= \left\lfloor\frac{(n-1)^2}{4}\right\rfloor,\\ \sum_{i=1}^{n}[2(n-i)+1]\left\lfloor\frac{i-1}{2}\right\rfloor &= \binom{n}{3} \end{aligned}$$

give the asserted lower bound. If n is even, both nodes are selected at every occupied level, so the minimizing subset is unique and its leading coefficient is nonzero. If n is odd, the two minimizing subsets differ only in the node chosen at the last level. Their coefficients have different degrees in $a(0)$, an affine indeterminate in μ, and hence cannot cancel identically. This proves the exact generic order. Since $V_n^{(r)}\in R$, the vanishing of its coefficients below this order also proves divisibility in R. □

2.3 Lowest coefficients and exceptional specializations

Let

$$L = (1+\alpha)\mu - 1, \qquad w_r = v_r|_{q=0} = \frac{(-1)^r L - \alpha^{r+1}}{1+\alpha}. \tag{2.8}$$

The expression for w_r is a polynomial after cancellation of the apparent denominator; alternatively it is determined by $w_0 = \mu - 1$ and $w_r = -w_{r-1} - \alpha^r$.

Proposition 2.2. *Writing $e = e_{n,r}$, the lowest coefficients are*

$$[q^e]V^{(r)}_{2m} = (-1)^{m(r+1)}\alpha^{m(2m+r-1)}L^m, \tag{2.9}$$

$$[q^e]V^{(r)}_{2m+1} = (-1)^{m(r+1)}\alpha^{m(2m+r+1)}L^m w_r. \tag{2.10}$$

Formula (2.9) *applies for $m \geq 1$ and* (2.10) *for $m \geq 0$.*

Proof. At every occupied level, the leading weight coefficients are $A = L/(1+\alpha)$ and $B = -\alpha/(1+\alpha)$. For $n = 2m$, the product of all weight coefficients is $(AB)^m$. The factors x_i^r contribute $(-1)^{mr}\alpha^{mr}$. Each same-level Vandermonde square contributes $(1+\alpha)^2$, cancelling the corresponding denominator in AB. For a positive node at level j, the higher levels contribute a further factor $\alpha^{4(m-1-j)}$ to the leading Vandermonde coefficient. Multiplication gives (2.9).

For $n = 2m+1$, both nodes at each level $0, \ldots, m-1$ are selected. The extra node contributes an additional α^{2m} through its interactions with the preceding positive nodes. The sum of its two possible weight-and-shift coefficients is $A(-1)^r + B\alpha^r = w_r$. This yields (2.10). □

Remark 2.3. For $\alpha = 2, \mu = 3, r = 2$, one has $w_2 = 0$. Thus at $n = 5$ the coefficient of the generic lowest power q^{18} vanishes. Exact polynomial calculation gives order 19 in this example (appendix B). This is an exceptional specialization, not a failure of the generic identity. No equality after every specialization is claimed here.

2.4 The highest degree

Proof of (1.6). Induction in (1.2) gives $\deg_q v_k = k(k+1)/2$ with leading coefficient $\mu - 1$. For a permutation π of $\{0, \ldots, n-1\}$, the corresponding term of the determinant has degree $\sum_i (i + \pi(i) + r)(i + \pi(i) + r + 1)/2$. The only part depending on the permutation is $\sum_i i\pi(i)$. Indeed, the difference between its value at the identity and at π is $\frac{1}{2}\sum_i (i - \pi(i))^2$, which is positive unless π is the identity. Thus the identity permutation uniquely supplies the highest power of q, whose coefficient is $(\mu - 1)^n$. This proves the degree formula. □

For fixed $t \geq 0$ and $r = \lfloor tn \rfloor$, these algebraic identities give

$$\frac{D_{n,r}}{n^3} \longrightarrow C(t), \qquad \frac{e_{n,r}}{n^3} \longrightarrow b_0(t) := \frac{1}{6} + \frac{t}{4}. \tag{2.11}$$

3 The prescribed resonant cyclotomic divisor

This section gives a self-contained construction of the prescribed divisor. The proof establishes a guaranteed divisor; it does not establish that all its cyclotomic exponents are exact.

Fix an integer $d \geq 1$ and put $h = q^d - 1$, $a = \alpha^d$. Let T_d denote the shift by d on sequences and define

$$W_{s,k} = (\mathsf{T}_d - a)^s v_k = \sum_{j=0}^{s} \binom{s}{j} (-a)^{s-j} v_{k+jd}. \tag{3.1}$$

We use the sans-serif symbol T_d for this operator, to distinguish it from the scalar sum $T_d(n,p)$ in (1.12).

Lemma 3.1 (Resonance and lifting)**.** *For $s \geq 1$ and $k \geq 1$,*

$$W_{s,k} = (q^k - 1)W_{s,k-1} + q^k \sum_{j=1}^{s} \binom{s}{j} h^j W_{s-j,k+jd-1}. \tag{3.2}$$

For all $s, k \geq 0$,

$$h^{\min(s,\lfloor k/d \rfloor)} \mid W_{s,k} \quad \text{in } R. \tag{3.3}$$

Proof. Substitution of (1.2) into (3.1) cancels the inhomogeneous terms because their sum is $-\alpha^k \sum_{j=0}^{s} \binom{s}{j}(-a)^{s-j}a^j = 0$. Writing $q^{k+jd} - 1 = (q^k - 1) + q^k((1+h)^j - 1)$, the remaining terms are reorganized by the commuting-operator identity

$$((1+h)\mathsf{T}_d - a)^s = ((\mathsf{T}_d - a) + h\mathsf{T}_d)^s.$$

This gives (3.2).

For (3.3), use induction on s, and then on k for fixed s. The claim is immediate when $s = 0$ or $k < d$. In the first term of (3.2), the floor $\lfloor k/d \rfloor$ either stays unchanged from $k-1$, or increases by one when $d \mid k$; in the latter case $h \mid q^k - 1$ supplies the additional factor. In a term with $j \geq 1$, the explicit power of h and the induction hypothesis give an exponent at least

$$j + \min\left(s - j, \left\lfloor \frac{k + jd - 1}{d} \right\rfloor\right) \geq \min\left(s, \left\lfloor \frac{k}{d} \right\rfloor + 2j - 1\right) \geq \min\left(s, \left\lfloor \frac{k}{d} \right\rfloor\right).$$

This completes both inductions. □

Proposition 3.2 (Shifted divisor)**.** *The polynomial $\mathcal{D}_{n,r}$ of* (1.14) *divides $V_n^{(r)}$ in R. For $r = 0$ it can also be written as*

$$\mathcal{D}_{n,0}(q) = q^{\binom{n}{3}} \prod_{\ell \geq 1} (q^\ell - 1)^{2(n-2\ell)_+}. \tag{3.4}$$

Proof. Let integers a_i, b_j satisfy

$$0 \leq a_i \leq \left\lfloor \frac{i}{d} \right\rfloor, \qquad 0 \leq b_j \leq \left\lfloor \frac{j}{d} \right\rfloor, \qquad a_i + b_j \leq \left\lfloor \frac{i+j+r}{2d} \right\rfloor. \tag{3.5}$$

Apply lower unitriangular row transformations

$$R_i \longmapsto \sum_{u=0}^{a_i} \binom{a_i}{u} (-\alpha^d)^{a_i - u} R_{i-(a_i-u)d}$$

and analogous column transformations with orders b_j. All indices are nonnegative by (3.5). The transformed entry is

$$W_{a_i+b_j,k_0}, \qquad k_0 = i + j + r - d(a_i + b_j) \geq d(a_i + b_j).$$

By Lemma 3.1, this entry is divisible by $h^{a_i+b_j}$. Factoring h^{a_i} from row i and h^{b_j} from column j gives $h^{\sum_i a_i + \sum_j b_j} \mid V_n^{(r)}$.

For $p + p' = r$, choose

$$a_i = \min\left\{\left\lfloor \frac{i}{d} \right\rfloor, \left\lfloor \frac{i+p}{2d} \right\rfloor\right\}, \qquad b_j = \min\left\{\left\lfloor \frac{j}{d} \right\rfloor, \left\lfloor \frac{j+p'}{2d} \right\rfloor\right\}.$$

The floor inequality $\lfloor x\rfloor+\lfloor y\rfloor\le\lfloor x+y\rfloor$ verifies (3.5). Thus $(q^d-1)^{T_d(n,p)+T_d(n,p')}\mid V_n^{(r)}$. In particular, $\Phi_d(q)^{\nu_d(n,r)}\mid V_n^{(r)}$. The cyclotomic polynomials are distinct monic irreducibles over $\mathbb{Q}$, so their product divides over $\mathbb{Q}[\alpha,\mu][q]$ and, by monic division, over R. It is coprime to q, and theorem 1.1 supplies the remaining q-power.

At $r=0$, $\nu_d(n,0)=2S_d(n)$ and $S_d(n)=\sum_{m\ge1}(n-2dm)_+$. The exponent of $\Phi_d(q)$ in the product on the right of (3.4) is therefore exactly $2S_d(n)$. □

In particular,

$$\deg_q\mathcal{D}_{n,0}=\begin{cases}n(n-1)^2/4, & n\text{ odd},\\ n^2(n-2)/4, & n\text{ even},\end{cases}\qquad \deg_q\mathcal{D}_{n,0}=\frac{1}{4}n^3+O(n^2). \tag{3.6}$$

This is the divisor form of the expectation recorded in [1, Remark 7]. The factors $(q^\ell-1)$ in (3.4) are not themselves pairwise coprime; reassembly through their irreducible cyclotomic factors is essential.

4 Positive moments and sharp analytic bounds

4.1 The product and the actual remainder

Comparison of coefficients in (1.1) gives

$$E_q(qz)=(1+z)E_q(z). \tag{4.1}$$

Iterating downward and using $E_q(0)=1$ yields, for $|q|>1$,

$$E_q(z)=\prod_{j\ge1}(1+zq^{-j}). \tag{4.2}$$

The product converges locally uniformly in z. In the remainder of this section assume $q>1,\alpha>0$ and $\mu=E_q(\alpha)$.

Proof of the moment and positivity assertions in theorem 1.2. Let $P_j=\prod_{\ell=1}^{j}(1+\alpha q^{-\ell})$ and $P_0=1$. The weights in (1.7) satisfy $w_j=P_j-P_{j-1}$, so telescoping gives $\sum_{j\ge1}w_j=\mu-1=v_0$. They also satisfy

$$w_1q=\alpha,\qquad w_{j+1}q^{j+1}-w_jq^j=\alpha w_j.$$

Since $0<w_j\le\alpha\mu q^{-j}$, all moment sums converge absolutely. The displayed relations, followed by an index shift, verify the recurrence for $\sum_{j\ge1}w_j(\alpha q^{-j})^k$. This proves the moment identity by uniqueness of the recurrence solution.

Finite Cauchy–Binet followed by the convergent limit now gives

$$V_n^{(r)}=\sum_{1\le j_1<\cdots<j_n}\prod_{a=1}^{n}w_{j_a}x_{j_a}^r\prod_{a<b}(x_{j_b}-x_{j_a})^2. \tag{4.3}$$

Each term is positive, and at least one exists. The finite determinants converge because their entries do. This proves strict positivity and the convergence of the nonnegative expansion. □

4.2 A two-sided product estimate

Proof of (1.9) *and* (1.10). Retain the subset $j_a = a$ in (4.3) and use $w_a \geq \alpha q^{-a}$. Since

$$x_{j_a} - x_{j_b} = \alpha q^{-j_a}(1 - q^{-(j_b - j_a)}) \quad (a < b),$$

this yields the lower bound in (1.9); its q-exponent is

$$\sum_{a=1}^{n} [r + 2(n-a) + 1]a = K_{n,r}.$$

For the upper bound, use $w_j \leq \alpha\mu q^{-j}$ and remove all the factors $(1 - q^{-(j_b - j_a)})^2 \leq 1$. It remains to sum

$$\sum_{j_1 < \cdots < j_n} q^{-\sum_{a=1}^{n} [r+2(n-a)+1] j_a} = q^{-K_{n,r}} \prod_{m=1}^{n} (1 - q^{-m(m+r)})^{-1}. \tag{4.4}$$

Indeed, write $j_a = a + g_1 + \cdots + g_a$, where each $g_a \geq 0$. The coefficient of g_a in the exponent is $(n - a + 1)(n - a + 1 + r)$, and the n sums are geometric. The total power of α is $n + nr + n(n-1) = n^2 + rn$. This proves both bounds.

The lower product is at least $\prod_{k \geq 1} (1 - q^{-k})^{2n}$. The upper reciprocal product is at most $\prod_{m \geq 1} (1 - q^{-m^2})^{-1}$, independent of r and n. Taking logarithms proves the uniform error in (1.10). For $r = \lfloor tn \rfloor$, division by $n^3 \log q$ gives $A(t) = 1/3 + t/2$. □

Corollary 4.1 (No faster cubic decay for the same positive determinant)**.** *For fixed $q > 1, \alpha > 0$ at $\mu = E_q(\alpha)$ and fixed $t \geq 0$,*

$$\lim_{n \to \infty} \frac{-\log V_n^{(\lfloor tn \rfloor)}}{n^3 \log q} = \frac{1}{3} + \frac{t}{2}.$$

Consequently no valid eventual estimate for this same sequence can replace this exponent by a larger fixed constant in a bound of the form $q^{-An^3 + o(n^3)}$.

This conclusion uses positivity and two-sided bounds. It is not an inference from a one-sided determinant estimate, and it is not asserted for arbitrary complex specializations.

5 The divisor-degree budget

Lemma 5.1 (A floor-sum inequality)**.** *For integers $L, n \geq 1$, define $G(p) = \sum_{i=0}^{n-1} \lfloor (i+p)/L \rfloor$ for $p \geq 0$. Then*

$$G(p) + G(p') \leq G(0) + G(p + p') \quad (p, p' \geq 0). \tag{5.1}$$

Consequently,

$$\nu_d(n, r) \leq S_d(n) + S_d(n + r) - S_d(r). \tag{5.2}$$

Proof. Write $n = aL + b$ with $0 \leq b < L$, and $p = cL + s$ with $0 \leq s < L$. Direct summation gives

$$G(p) - G(0) = ap + bc + (b + s - L)_+.$$

The last two terms form $H(p) = b\lfloor p/L \rfloor + (b + (p \bmod L) - L)_+$. Consider the periodic binary sequence with $L - b$ consecutive zeros followed by b consecutive ones in each period. The minimum number of ones in an interval of length p is exactly $H(p)$: complete periods contribute bc, and a residual interval of length s contributes at least $(s - (L - b))_+$, a minimum attainable by placing it across the zero block. Splitting an arbitrary interval of length $p + p'$

into pieces of lengths p, p' shows that its number of ones is at least $H(p) + H(p')$. Taking a minimum proves superadditivity of H, hence (5.1).

Take $L = 2d$. Since $T_d(n,p) \le G(p)$, for $p + p' = r$ one obtains

$$T_d(n,p) + T_d(n,p') \le G(0) + G(r) = S_d(n) + S_d(n+r) - S_d(r).$$

Maximization over the split proves (5.2). □

Proposition 5.2. *For every n, r,*

$$\deg_q \mathcal{D}_{n,r} \le e_{n,r} + F(n) + F(n+r) - F(r), \tag{5.3}$$

where

$$F(N) := \sum_{d\ge1} \varphi(d) S_d(N) = \sum_{\ell\ge1} \ell(N-2\ell)_+ = \frac{N^3}{24} + O(N^2). \tag{5.4}$$

In particular, for $r = \lfloor tn \rfloor$ and fixed $t \ge 0$,

$$\limsup_{n\to\infty} \frac{\deg_q \mathcal{D}_{n,r}}{n^3} \le \frac{1}{4} + \frac{3t}{8} + \frac{t^2}{8}. \tag{5.5}$$

Proof. Use $S_d(N) = \sum_{m\ge1}(N-2dm)_+$ and $\sum_{d|\ell} \varphi(d) = \ell$ to obtain the middle expression in (5.4). Summing the resulting quadratic polynomial in ℓ gives its asymptotic. Multiply (5.2) by $\varphi(d)$ and sum over d to prove (5.3). Since

$$\frac{1 + (1+t)^3 - t^3}{24} = \frac{1}{12} + \frac{t}{8} + \frac{t^2}{8},$$

combination with $b_0(t) = 1/6 + t/4$ proves (5.5). □

Remark 5.3 (Which divisor is bounded). The right side of (5.3) bounds the degree of the *specified* polynomial (1.14). It does not bound the degree of every polynomial dividing $V_n^{(r)}$, nor the full cyclotomic valuation of a specialized determinant. Replacing this distinction by an assertion of universal sharpness would not be justified by the proof.

6 Arithmetic comparison and shift optimality

For positive rational $q = \rho/\sigma > 1$ in lowest terms, with $\rho > \sigma \ge 1$, define

$$\gamma_q = \frac{\log \rho}{\log \sigma} \quad (\sigma > 1), \qquad \gamma_q = \infty \quad (\sigma = 1). \tag{6.1}$$

The symbol γ_q is a height ratio; it is not Euler's constant. We give the norm calculation explicitly to specify the comparison whose optimality is at issue. It is the shifted counterpart of [1, Section 5].

6.1 Denominator clearing and conjugates

For fixed $q > 1$, the cyclotomic product satisfies

$$\log |\mathcal{D}_{n,r}(q)| = (\deg_q \mathcal{D}_{n,r}) \log q + O_q(n^2 \log(n+1)). \tag{6.2}$$

To see this, use the Möbius product formula to bound $|\log|\Phi_d(q)| - \varphi(d)\log q|$ by $\sum_{m\ge1} |\log(1-q^{-m})|$, a finite constant independent of d. Also $\nu_d(n,r) \le 2\sum_{i<n} \lfloor i/d \rfloor \le n^2/d$, and $\nu_d(n,r) = 0$ for $d \ge n$. Summing proves (6.2).

Suppose that $\alpha > 0$ and $\mu = E_q(\alpha)$ are algebraic, and put $K = \mathbb{Q}(\alpha, \mu)$, $D = [K : \mathbb{Q}]$. Choose an integer $c \geq 1$ such that $c\alpha$ and $c\mu$ are algebraic integers. The polynomial bounds

$$\deg_\alpha V_n^{(r)} \leq n(n-1+r), \qquad \deg_\mu V_n^{(r)} \leq n$$

follow from the recurrence and determinant expansion. They remain valid for the quotient by the monic q-only polynomial $\mathcal{D}_{n,r}$. Writing $b_{n,r} = \deg_q \mathcal{D}_{n,r}$, it follows that

$$I_{n,r} := c^{n(n+r)} \sigma^{D_{n,r}-b_{n,r}} \frac{V_n^{(r)}(q,\alpha,\mu)}{\mathcal{D}_{n,r}(q)} \quad \text{is an algebraic integer.} \tag{6.3}$$

It is nonzero by theorem 1.2.

For any embedding $\iota : K \hookrightarrow \mathbb{C}$, solving the recurrence as in (1.4), without imposing the actual-value identity on $\iota(\mu)$, gives a bound

$$|\iota(v_k)| \leq C_\iota q^{k(k+1)/2}.$$

Indeed, the corresponding finite partial sums of $\sum \iota(\alpha)^j / \prod_{\ell=1}^{j}(q^\ell - 1)$ are bounded. The determinant expansion and the same degree maximization as in theorem 1.1 therefore give

$$|\iota(V_n^{(r)})| \leq n!\, C_\iota^n q^{D_{n,r}}. \tag{6.4}$$

At the distinguished real embedding the much stronger estimate (1.10) applies.

Proposition 6.1 (The cubic norm criterion)**.** *Fix $t \geq 0$ and $r = \lfloor tn \rfloor$. Along any infinite subsequence on which $b_{n,r}/n^3 \to B$, the assumption $[\mathbb{Q}(\alpha,\mu) : \mathbb{Q}] = D$ is contradicted if*

$$A(t) + C(t) - D(C(t) - B) > 0, \qquad \gamma_q > \frac{A(t) + C(t)}{A(t) + C(t) - D(C(t) - B)}. \tag{6.5}$$

For integer q, the second condition is automatic when the first holds.

Proof. The logarithm of the c-factor in (6.3) is $O(n^2)$ because $r = O(n)$. Multiply the distinguished bound for $I_{n,r}$ by the $D-1$ bounds arising from (6.4), and use (6.2). The result is

$$\log |N_{K/\mathbb{Q}}(I_{n,r})| \leq \big((A + C)\log\sigma - (A + C - D(C - B))\log\rho\big)n^3 + o(n^3). \tag{6.6}$$

The coefficient is strictly negative under (6.5). But the norm is a nonzero integer, whose absolute value is at least one. This is a contradiction along the subsequence. □

Corollary 6.2 (Unshifted consequences in the positive range)**.** *Let $q = \rho/\sigma > 1$ be rational and $\alpha > 0$ algebraic. If $\gamma_q > 6$, then α and $E_q(\alpha)$ cannot both lie in a number field of degree at most two. In particular, for positive rational α, $E_q(\alpha)$ is neither rational nor quadratic algebraic. If $\alpha > 0$ is rational and $\gamma_q > 12/7$, then $E_q(\alpha)$ is irrational.*

Proof. At $t = 0$, (3.6) gives $B = 1/4$ and $A = 1/3, C = 2/3$. Substitution of $D = 1, 2$ into (6.5) gives $12/7$ and 6, respectively. □

These known arithmetic values agree with [2, Corollary 1.1]. The positive-range corollary is proved directly here to identify exactly which norm comparison is subject to the shift barrier. The more general complex-parameter results in [1, 2] are not needed for this positivity argument.

6.2 The global shift barrier

Proof of theorem 1.3. Divisibility is Proposition 3.2; the constants A, C follow from theorems 1.1 and 1.2; and Proposition 5.2 gives $B \le \overline{B}(t)$. Since $A+B>0$ and the ratio decreases with B,

$$\frac{A+C}{A+B} \ge \frac{A+C}{A+\overline{B}(t)} = \frac{24+36t+12t^2}{14+21t+3t^2}.$$

The exact difference from the unshifted value is

$$\frac{24+36t+12t^2}{14+21t+3t^2} - \frac{12}{7} = \frac{48t^2}{7(14+21t+3t^2)}. \tag{6.7}$$

This is nonnegative, and positive for $t>0$. For degree two, set $\eta = (A+B)/(A+C)$. The preceding inequality implies $\eta \le 7/12$. When $2\eta - 1 > 0$,

$$\tau_2 = \frac{1}{2\eta - 1} \ge 6.$$

Otherwise the cubic comparison gives no finite degree-two critical value. Finally, (3.6) shows that the bounds are attained at $t=0$. This proves the theorem, including its accumulation-point formulation. □

Remark 6.3 (What the theorem does not rule out). The exact analytic exponent is established for $q>1, \alpha>0$ at the actual value, and the barrier uses precisely the divisor (1.14) and the conjugate bound (6.4). It is not a universal theorem about irrationality thresholds. In particular, it leaves open extra specialization-dependent content, a larger uniform divisor, different minors or determinants, other conjugate bounds, and other denominator clearings. It neither determines every shifted cyclotomic valuation nor excludes a future general-parameter result below 12/7.

7 Entire normalization and exact forced zeros

In this section and the subsequent analytic sections, fix a real number $q>1$ and put $p = q^{-1}$. The parameter α is now allowed to range over $\mathbb{C}$. Write

$$\pi_j = (p;p)_j = \prod_{\ell=1}^{j}(1-p^\ell), \qquad P_\infty = (p;p)_\infty, \qquad \mathcal{M} = \prod_{\ell\ge 1}(1-p^\ell)^{-\ell}. \tag{7.1}$$

As usual, $(a;p)_j = \prod_{\ell=0}^{j-1}(1-ap^\ell)$, and an empty product is one. Set

$$H_{n,r}(\alpha) = V_n^{(r)}(q,\alpha,E_q(\alpha)), \tag{7.2}$$

$$\mathcal{A}_{n,r} = \prod_{i,j=0}^{n-1}(1-p^{r+i+j+1})^{-1}, \tag{7.3}$$

$$\mathcal{C}_{n,r} = \det\left(\frac{p^{r+i+j+1}}{1-p^{r+i+j+1}}\right)_{i,j=0}^{n-1} = p^{K_{n,r}}\prod_{\ell=1}^{n-1}(1-p^\ell)^{2(n-\ell)}\mathcal{A}_{n,r}, \tag{7.4}$$

$$\mathcal{L}_{n,r}(\alpha) = \alpha^{n^2+rn}\prod_{m=1}^{n-1}(1+\alpha p^m)^{n-m}, \tag{7.5}$$

where $K_{n,r}$ is as in theorem 1.2. The evaluation in (7.4) is the Cauchy determinant formula; in particular, $\mathcal{C}_{n,r}>0$.

Theorem 7.1 (Exact forced zeros)**.** *For $n \geq 1$ and $r \geq 0$, the function $H_{n,r}$ is entire and*

$$\operatorname{ord}_{\alpha=0} H_{n,r} = n^2 + rn, \qquad \operatorname{ord}_{\alpha=-p^{-m}} H_{n,r} = n - m \quad (1 \leq m < n). \tag{7.6}$$

Consequently, the quotient

$$\Psi_{n,r}(\alpha) = \frac{H_{n,r}(\alpha)}{\mathcal{L}_{n,r}(\alpha)\mathcal{C}_{n,r}} \tag{7.7}$$

extends to an entire function, and $\Psi_{n,r}(0) = 1$.

Proof. The recurrence defining v_k shows directly that $H_{n,r}$ is entire. The moment representation from section 4 extends to complex α:

$$\begin{aligned} v_k &= \sum_{j\geq 1} w_j x_j^k, \qquad x_j = \alpha p^j, \qquad w_j = \alpha p^j \prod_{\ell=1}^{j-1}(1+\alpha p^\ell), \\ H_{n,r}(\alpha) &= \sum_{1\leq j_1<\cdots<j_n} \prod_{a=1}^{n} w_{j_a} x_{j_a}^r \prod_{a<b}(x_{j_b} - x_{j_a})^2. \end{aligned} \tag{7.8}$$

Indeed, both series converge normally on compact subsets of $\mathbb{C}$. For example, on $|\alpha| \leq R$ the partial products in w_j are at most $\exp(Rp/(1-p))$ in absolute value. The remaining geometric factors give an absolutely summable bound for the Cauchy–Binet series. The moment identity follows either from the recurrence or from the identity theorem applied to the positive real axis.

At $\alpha = 0$, extracting the powers of α in (7.8) gives

$$H_{n,r}(\alpha) = \alpha^{n^2+rn}(\mathcal{C}_{n,r} + O(\alpha)).$$

Since $\mathcal{C}_{n,r} > 0$, this proves the first assertion of (7.6) and the value of $\Psi_{n,r}(0)$.

Fix $m < n$ and let $\alpha_0 = -p^{-m}$. The first m weights in (7.8) are nonzero at α_0; every subsequent weight has a simple zero there. Thus every summand has order at least $n - m$. Terms of exactly this order must contain all of the first m nodes. There is no cancellation among their leading coefficients:

$$\operatorname{sgn} w_j(\alpha_0) = (-1)^j \quad (j \leq m), \qquad \operatorname{sgn} w_j'(\alpha_0) = (-1)^m \quad (j > m).$$

The factors x_j^r contribute the common sign $(-1)^{rn}$, and the Vandermonde squares are strictly positive. At least one such summand exists. Normal convergence permits coefficient extraction, so the sum of these nonzero coefficients is nonzero. This proves the second assertion and removes every apparent pole in (7.7). □

8 A Schur expansion and a uniform scaling limit

We use the standard Schur identities in [3, Chapter I]. For a partition λ, let $|\lambda|$ and $\ell(\lambda)$ denote its size and length. A cell $\square = (i, b)$ in its diagram has content $c(\square) = b - i$ and hook length $h(\square)$, and $b(\lambda) = \sum_i (i-1)\lambda_i$. Put

$$X_n = (1, p, \ldots, p^{n-1}), \qquad X = (1, p, p^2, \ldots).$$

The principal specializations are

$$s_\lambda(X) = \frac{p^{b(\lambda)}}{\prod_{\square\in\lambda}(1 - p^{h(\square)})}, \qquad s_\lambda(X_n) = s_\lambda(X) \prod_{\square\in\lambda}(1 - p^{n+c(\square)}) \quad (\ell(\lambda) \leq n), \tag{8.1}$$

and $s_\lambda(X_n) = 0$ for $\ell(\lambda) > n$.

Proposition 8.1 (Exact partition expansion)**.** *Define*

$$F_{n,r}(z) = \Psi_{n,r}(p^{-n-r}z), \qquad a = p^r. \tag{8.2}$$

Then, for every $z \in \mathbb{C}$,

$$F_{n,r}(z) = \frac{1}{\mathcal{A}_{n,r}} \sum_{\lambda} p^{|\lambda|} s_\lambda(X_n)^2 \prod_{\square \in \lambda} (a + zp^{c(\square)}). \tag{8.3}$$

The series converges normally on compact subsets of $\mathbb{C}$. Its normalizing factor satisfies

$$\mathcal{A}_{n,r} = \sum_{\lambda} p^{(r+1)|\lambda|} s_\lambda(X_n)^2. \tag{8.4}$$

Proof. For a configuration $j_1 < \cdots < j_n$ in (7.8), write $j_a = a + \lambda_{n-a+1}$. The alternant definition of Schur polynomials gives the Vandermonde ratio $s_\lambda(X_n)$. Relative to the ground configuration $j_a = a$, the geometric weights and the squared Vandermonde therefore contribute $p^{(r+1)|\lambda|} s_\lambda(X_n)^2$. The remaining product is

$$\prod_{a=1}^{n} \prod_{\ell=a}^{j_a-1} (1 + \alpha p^\ell) = \prod_{\square \in \lambda} (1 + \alpha p^{n+c(\square)}). \tag{8.5}$$

Ordinary Cauchy summation gives (8.4). Substitution of $\alpha = p^{-n-r}z$ in (8.5), together with $p^{r|\lambda|}$ from the weights, gives (8.3). Normal convergence also follows from the summable majorant established in the next lemma. □

For $0 \le a \le 1$, define

$$\mathcal{A}_\infty(a) = \prod_{\ell \ge 1} (1 - ap^\ell)^{-\ell}, \qquad \Phi_p(z) = \prod_{\ell \ge 1} (1 + zp^\ell)^\ell. \tag{8.6}$$

The second product is entire, and $\mathcal{A}_\infty(1) = \mathcal{M}$. We also write

$$\mathcal{A}_{\infty,r} = \mathcal{A}_\infty(p^r) = \prod_{\ell \ge 1} (1 - p^{r+\ell})^{-\ell}. \tag{8.7}$$

Lemma 8.2 (Summation and a content moment)**.** *For $0 \le a \le 1$ and $z \in \mathbb{C}$,*

$$\sum_{\lambda} p^{|\lambda|} s_\lambda(X)^2 \prod_{\square \in \lambda} (a + zp^{c(\square)}) = \mathcal{A}_\infty(a) \Phi_p(z). \tag{8.8}$$

The convergence is locally uniform in z and uniform in $a \in [0,1]$. Furthermore, if $C_1(\lambda) = \sum_{\square \in \lambda} p^{c(\square)}$, then for $R \ge 0$

$$\sum_{\lambda} p^{|\lambda|} s_\lambda(X)^2 C_1(\lambda) \prod_{\square \in \lambda} (1 + Rp^{c(\square)}) = \mathcal{M} \Phi_p(R) \frac{(1+R)p}{(1-p)^2}. \tag{8.9}$$

Proof. Use the supersymmetric specialization $\mathsf{A}_{a,z} = aX + \omega(zX)$, where ω is the standard involution on symmetric functions. Equivalently, its power sums are

$$p_k[\mathsf{A}_{a,z}] = \frac{a^k + (-1)^{k-1} z^k}{1 - p^k}.$$

The hook-content specialization gives

$$s_\lambda[\mathsf{A}_{a,z}] = s_\lambda(X) \prod_{\square \in \lambda} (a + zp^{c(\square)}). \tag{8.10}$$

For $a > 0$ this follows by homogeneity from the usual specialization $p_k \mapsto (1-u^k)/(1-p^k)$ with $u = -z/a$; the case $a = 0$ follows polynomially.

Here it is important to justify Cauchy summation for arbitrarily large $|z|$. First take finite alphabets aX_N, RX_N, and pX_N, with $R \geq 0$. The mixed ordinary and dual Cauchy identity gives

$$\sum_{\lambda} s_\lambda[aX_N + \omega(RX_N)]s_\lambda(pX_N) = \prod_{i,j=0}^{N-1} \frac{1+Rp^{i+j+1}}{1-ap^{i+j+1}}.$$

All terms on the left are nonnegative, by the skew-Schur expansion of a supersymmetric Schur function. Increasing N and using monotone convergence proves that the corresponding infinite sum equals $\mathcal{A}_\infty(a)\Phi_p(R)$, which is finite for every R. For complex z with $|z| \leq R$ and $a \in [0,1]$, the absolute value of each summand in (8.8) is bounded by the summand with $(a,z) = (1,R)$. This proves normal convergence and the identity by analytic continuation. In particular, no logarithmic power series with a restricted z-radius is being used.

To obtain (8.9), apply the same Cauchy identity to $\mathsf{A}_{1,R}$ and $p\mathsf{A}_{1,\varepsilon}$. For sufficiently small nonnegative ε it gives

$$\sum_{\lambda} p^{|\lambda|} s_\lambda(X)^2 \prod_{\square\in\lambda} (1+Rp^{c(\square)})(1+\varepsilon p^{c(\square)}) = \mathcal{M}\Phi_p(R)\Phi_p(\varepsilon) \prod_{\ell\geq 1} (1-R\varepsilon p^\ell)^{-\ell}.$$

Taking the coefficient of ε proves the assertion, since $\sum_{\ell\geq 1} \ell p^\ell = p/(1-p)^2$. Coefficient extraction is justified either by the preceding finite-alphabet argument and nonnegativity, or by the convergent identity in a neighborhood of zero. □

Theorem 8.3 (Uniform shifted scaling limit)**.** *For every $R \geq 0$, every $n \geq 1$, and every integer $r \geq 0$,*

$$\sup_{|z|\leq R} |F_{n,r}(z) - \Phi_p(z)| \leq \frac{2\mathcal{M}\Phi_p(R)(R+2)}{(1-p)^2}\, p^{n+1}. \tag{8.11}$$

Thus $F_{n,r} \to \Phi_p$ locally uniformly on $\mathbb{C}$, uniformly in the entire range $r \geq 0$. The same conclusion, with an $O_{p,R,d}(p^n)$ bound, holds for every fixed derivative order d on $|z| \leq R$.

Proof. Let $t_{\lambda,n} = s_\lambda(X_n)/s_\lambda(X)$. By (8.1), $0 \leq t_{\lambda,n} \leq 1$ and

$$1 - t_{\lambda,n}^2 \leq 2p^n C_1(\lambda). \tag{8.12}$$

When $\ell(\lambda) \leq n$, this is the elementary inequality $1 - \prod(1-u_j)^2 \leq 2\sum u_j$ applied to $u_j = p^{n+c(\square)} \in (0,1)$. When $\ell(\lambda) > n$, the cell $(n+1,1)$ gives $p^n C_1(\lambda) \geq 1$, and the inequality remains true.

For $0 \leq a \leq 1$, denote the numerator of (8.3) by $N_{n,a}(z)$ and its infinite-alphabet version by $N_{\infty,a}(z)$. Equations (8.12) and (8.9) give, uniformly for $|z| \leq R$,

$$|N_{n,a}(z) - N_{\infty,a}(z)| \leq \frac{2\mathcal{M}\Phi_p(R)(1+R)}{(1-p)^2}\, p^{n+1}.$$

Writing $\mathcal{A}_n(a) = \prod_{i,j<n}(1-ap^{i+j+1})^{-1}$, the same estimate with $R = 0$ gives

$$|\mathcal{A}_n(a) - \mathcal{A}_\infty(a)| \leq \frac{2\mathcal{M}}{(1-p)^2}\, p^{n+1}.$$

Since $\mathcal{A}_n(a) \geq 1$, $|\Phi_p(z)| \leq \Phi_p(R)$, and $N_{\infty,a} = \mathcal{A}_\infty(a)\Phi_p$, division proves (8.11) for $a = p^r$. Cauchy's derivative estimates on a slightly larger disk prove the derivative assertion. □

Corollary 8.4 (Zero clusters and the closest remaining zero)**.** *Fix $k \geq 1$, and choose a sufficiently small disk centered at $-p^{-k}$ that contains no other zero of Φ_p. For all sufficiently large n, independently of $r \geq 0$, this disk contains exactly k zeros of $F_{n,r}$, counted with multiplicity. All these zeros tend to $-p^{-k}$, uniformly in r.*

In particular, for all sufficiently large n, each $\Psi_{n,r}$ has a unique zero of smallest modulus, denoted $\beta_{n,r}$. It is simple, real, and negative, and

$$\sup_{r\geq 0} |p^{n+r}\beta_{n,r} + p^{-1}| \longrightarrow 0. \tag{8.13}$$

Moreover, $\Psi_{n,r} \to 1$ locally uniformly in the unscaled α plane, uniformly in r. Hence, on every fixed bounded disk, the zeros of $H_{n,r}$ are eventually exactly those in (7.6)*, with the stated multiplicities.*

Proof. The product in (8.6) has precisely the zeros $-p^{-k}$, of multiplicity k. Apply Rouché's theorem to the boundary of each chosen disk, using theorem 8.3. The same argument on $|z| = \rho$, with $p^{-1} < \rho < p^{-2}$, shows that there is exactly one zero in that disk. The zero near $-p^{-1}$ is therefore the unique closest one. It is simple and real because the functions have real coefficients and the disk is invariant under complex conjugation. Its limit is negative, proving the sign assertion and (8.13).

Finally, $\Psi_{n,r}(\alpha) = F_{n,r}(\alpha p^{n+r})$ and $\Phi_p(0) = 1$. The quantitative theorem therefore gives convergence to one uniformly on fixed α-disks and uniformly in r. Combining this with theorem 7.1 proves the last assertion. □

Remark 8.5. For $k > 1$, the cluster statement alone does not assert that the k zeros remain a multiple zero or that they are real. The splitting and reality questions require additional estimates; the double cluster is analyzed below.

8.1 The full unshifted leading constant and its first correction

Theorem 8.6 (Fixed-parameter refinement)**.** *For each $R > 0$, uniformly on $|\alpha| \leq R$,*

$$\frac{H_{n,0}(\alpha)}{p^{K_{n,0}} P_\infty^{2n} \mathcal{L}_{n,0}(\alpha)} = \mathcal{M}^3 \left(1 + \frac{(\alpha-4)p^{n+1}}{(1-p)^2} + O_{p,R}(p^{2n})\right). \tag{8.14}$$

At the forced zeros the quotient is understood by its entire extension. In particular, the first correction vanishes at $\alpha = 4$.

Proof. Write $F_n = F_{n,0}$. We have $F_n(0) = 1$ exactly. By the derivative part of theorem 8.3,

$$F_n'(0) = \Phi_p'(0) + O_p(p^n) = \frac{p}{(1-p)^2} + O_p(p^n),$$

and F_n'' is uniformly bounded on each fixed disk. Taylor expansion at zero therefore gives

$$\Psi_{n,0}(\alpha) = F_n(\alpha p^n) = 1 + \frac{\alpha p^{n+1}}{(1-p)^2} + O_{p,R}(p^{2n}). \tag{8.15}$$

The exact Cauchy evaluation can also be written as

$$\mathcal{C}_{n,0} = p^{K_{n,0}} \frac{\prod_{j=1}^{n-1} \pi_j^3}{\prod_{j=n}^{2n-1} \pi_j}.$$

Taking real logarithms of its positive factors yields the exact identity

$$\log \frac{\mathcal{C}_{n,0}}{p^{K_{n,0}} P_\infty^{2n} \mathcal{M}^3} = 4 \sum_{m\geq 1} m \log(1 - p^{n+m}) - \sum_{m\geq 1} m \log(1 - p^{2n+m}).$$

Consequently,

$$\frac{\mathcal{C}_{n,0}}{p^{K_{n,0}}P_\infty^{2n}\mathcal{M}^3} = 1 - \frac{4p^{n+1}}{(1-p)^2} + O_p(p^{2n}).$$

Multiplying this estimate by (8.15) proves (8.14). □

Corollary 8.7 (Unnormalized complex-parameter asymptotic)**.** *Fix* $\alpha \in \mathbb{C} \setminus \big(\{0\} \cup \{-p^{-m} : m \geq 1\}\big)$*, and put*

$$\mathcal{R}_p(\alpha) = \prod_{m\geq 1} (1+\alpha p^m)^{-m}.$$

Then

$$\begin{aligned} H_{n,0}(\alpha) = \alpha^{n^2} p^{K_{n,0}} \big(E_q(\alpha) P_\infty^2\big)^n \mathcal{M}^3 \mathcal{R}_p(\alpha) \\ \times \left(1 + \frac{2(\alpha-2)p^{n+1}}{(1-p)^2} + O_{p,\alpha}(p^{2n})\right). \end{aligned} \tag{8.16}$$

Proof. The absolutely convergent products give the exact identity

$$\frac{\mathcal{L}_{n,0}(\alpha)}{\alpha^{n^2} E_q(\alpha)^n \mathcal{R}_p(\alpha)} = \prod_{s\geq 1} (1+\alpha p^{n+s})^s = 1 + \frac{\alpha p^{n+1}}{(1-p)^2} + O_{p,\alpha}(p^{2n}).$$

Multiplication by (8.14) proves the claim. □

These are analytic refinements of the determinant estimates; they do not alter the cubic arithmetic comparison or assert a smaller irrationality threshold.

9 A classical moment model and the splitting of a double limiting zero

The specialization $\mu = 0$ admits a classical orthogonal-polynomial description. We use the q-Meixner formulas of Koekoek and Swarttouw [4, §3.13]; the identification below also specifies all normalizations needed for the determinant calculations. Throughout this section $q > 1$, $p = q^{-1}$, and

$$(a;p)_j = \prod_{s=0}^{j-1}(1-ap^s), \qquad \pi_j = (p;p)_j, \qquad P_\infty = (p;p)_\infty, \qquad \mathcal{M} = \prod_{s=1}^{\infty}(1-p^s)^{-s}. \tag{9.1}$$

All empty products equal 1.

9.1 The zero-value moment functional

Define the linear functional L_α on polynomials by

$$L_\alpha[x^s] = -v_s(q,\alpha,0), \qquad s \geq 0. \tag{9.2}$$

Its moments are polynomials in α and satisfy

$$m_0 = 1, \qquad m_s = (q^s-1)m_{s-1} + \alpha^s \quad (s \geq 1). \tag{9.3}$$

Lemma 9.1 (The classical q-Meixner specialization)**.** *For $\alpha > 0$, the functional L_α has the positive discrete representation*

$$L_\alpha[f] = \sum_{s=0}^{\infty} \lambda_s f(\alpha p^{-s}), \qquad \lambda_s = \frac{\alpha^{-s} p^{\binom{s}{2}}}{(-\alpha^{-1};p)_{s+1}}. \tag{9.4}$$

Its monic orthogonal polynomials are

$$\begin{aligned} P_j^\alpha(x) &= (-1)^j \pi_j p^{-j^2} M_j(x/\alpha; 1, \alpha^{-1}; p) \\ &= (-1)^j \pi_j p^{-j^2} \sum_{\ell=0}^{j} \frac{(p^{-j};p)_\ell (x/\alpha;p)_\ell}{\pi_\ell^2} (-\alpha p^{j+1})^\ell. \end{aligned} \tag{9.5}$$

Here M_j is the q-Meixner polynomial in the normalization of [4, §3.13]. Their squared norms are

$$\begin{aligned} h_j(\alpha) := L_\alpha[(P_j^\alpha)^2] &= \pi_j^2 p^{-j(2j+1)} (-\alpha p; p)_j \\ &= q^{\binom{j}{2}} \prod_{a=1}^{j} (q^a - 1)^2 \prod_{a=1}^{j} (\alpha + q^a). \end{aligned} \tag{9.6}$$

The polynomials in (9.5) *extend polynomially to $\alpha = 0$. The identities $L_\alpha[P_i^\alpha P_j^\alpha] = \delta_{ij} h_j(\alpha)$ hold for every complex α, with possible degeneracy when a norm vanishes.*

Proof. Put $t_s = \alpha^{-s} p^{\binom{s}{2}} / (-\alpha^{-1};p)_s$. Then $\lambda_s = t_s - t_{s+1}$, $t_0 = 1$, and $t_s \to 0$. Thus $\sum_s \lambda_s = 1$. All polynomial moments in (9.4) converge absolutely. Moreover,

$$(\alpha + 1)\lambda_0 = \alpha, \qquad (1 + \alpha p^{-s})\lambda_s = q\lambda_{s-1} \quad (s \geq 1).$$

Multiplication by $(\alpha p^{-s})^{k-1}$ and summation show that these moments satisfy (9.3). This proves (9.4).

The standard q-Meixner orthogonality formula, specialized to base p, $b = 1$, and $c = \alpha^{-1}$, reads in this probability normalization

$$\sum_{s=0}^{\infty} \lambda_s M_i(p^{-s}; 1, \alpha^{-1}; p) M_j(p^{-s}; 1, \alpha^{-1}; p) = \delta_{ij} p^{-j} (-\alpha p; p)_j.$$

The term $\ell = j$ in the terminating series shows that its leading coefficient as a polynomial in x is $(-1)^j p^{j^2}/\pi_j$. The multiplier in (9.5) therefore makes it monic, and multiplication of the displayed norm by $\pi_j^2 p^{-2j^2}$ gives (9.6). Finally, every expression $\alpha^\ell (x/\alpha;p)_\ell$ is a polynomial in (x, α). Both sides of each orthogonality identity are consequently polynomials in α. Their validity for $\alpha > 0$ extends them to all complex α. □

Proposition 9.2 (Exact zero-value determinant)**.** *For every $n \geq 1$,*

$$V_n(q, \alpha, 0) = (-1)^n q^{\binom{n}{3}} \prod_{d=1}^{n-1} (q^d - 1)^{2(n-d)} \prod_{m=1}^{n-1} (\alpha + q^m)^{n-m}. \tag{9.7}$$

In particular, its degree in α is $\binom{n}{2}$.

Proof. Changing from monomials to monic orthogonal polynomials is a triangular change of basis with determinant 1. Thus the Hankel determinant of L_α is $\prod_{j=0}^{n-1} h_j(\alpha)$. The moments of $V_n(q, \alpha, 0)$ are the negatives of those of L_α, so its determinant is $(-1)^n \prod_{j=0}^{n-1} h_j(\alpha)$. Substitution of (9.6), followed by collecting each factor, gives (9.7) for $\alpha > 0$. The polynomial identity extends to every α. □

We shall also use the fixed positive functional

$$B = L_0, \qquad B[x^s] = \prod_{a=1}^{s} (q^a - 1) = p^{-s(s+1)/2}\pi_s. \tag{9.8}$$

It is positive definite on real polynomials: at $\alpha = 0$ the polynomials P_j^0 remain monic and orthogonal, and their norms $\pi_j^2 p^{-j(2j+1)}$ are strictly positive.

9.2 Logarithmic derivatives at the product zeros

The following estimates concern the derivative in the independent variable μ, before the specialization $\mu = E_q(\alpha)$. We write a logarithmic derivative as a quotient of derivatives, so that no choice of a complex logarithm is needed.

Theorem 9.3. *Fix an integer $k \geq 1$. For $n \geq 1$ set $N = n + k$ and $\alpha_N = -p^{-N}$. Then $V_n(q, \alpha_N, 0) \neq 0$, and*

$$T_{n,k} := \frac{\partial_\mu V_n(q, \alpha_N, 0)}{V_n(q, \alpha_N, 0)} \sim \frac{(-1)^n}{P_\infty \pi_{k-1}} p^{-\{n^2 + (2k-3)n - 2k + 2\}/2}. \tag{9.9}$$

Proof. Solving the homogeneous part of the recurrence for v_s gives

$$v_s(q, \alpha, \mu) = -L_\alpha[x^s] + \mu B[x^s].$$

For fixed $\alpha = \alpha_N$, pass in this moment matrix to the monic basis $P_0^\alpha, \ldots, P_{n-1}^\alpha$. At $\mu = 0$ the matrix is diagonal with entries $-h_j(\alpha)$. These entries are nonzero because $j < n < N$. Differentiation of the determinant therefore gives the exact identity

$$T_{n,k} = -\sum_{j=0}^{n-1} \frac{B[(P_j^{\alpha_N})^2]}{h_j(\alpha_N)}. \tag{9.10}$$

Put $r = N - j$ and

$$A_j = (-1)^j \alpha_N^j p^{-\binom{j}{2}}, \qquad R_{j,N}(x) = P_j^{\alpha_N}(x)/A_j.$$

Reversing the terminating sum in (9.5), using $(p^{-j}; p)_s = (-1)^s p^{-js + \binom{s}{2}} \pi_j / \pi_{j-s}$, gives the exact formula

$$R_{j,N}(x) = \sum_{\ell=0}^{j} \frac{(-1)^\ell p^{r\ell + \binom{\ell}{2}}}{\pi_\ell} \left(\frac{\pi_j}{\pi_{j-\ell}}\right)^2 (-xp^N; p)_{j-\ell}. \tag{9.11}$$

We next prove both a limit and a bound for its B-norm.

For $0 \leq s \leq N - 2$, write $U_s(x) = (-xp^N; p)_s$. The finite p-binomial formula gives

$$[x^u]U_s(x) = p^{Nu + \binom{u}{2}} \frac{\pi_s}{\pi_u \pi_{s-u}} \leq P_\infty^{-1} p^{Nu + \binom{u}{2}} \quad (0 \leq u \leq s).$$

All its coefficients are nonnegative. In the product of its x^u and x^v terms, the moment bound from (9.8) gives the exponent

$$N(u+v) + \binom{u}{2} + \binom{v}{2} - \frac{(u+v)(u+v+1)}{2} = (N-1)(u+v) - uv.$$

Since $u, v \leq N-2$, $uv \leq (N-2)(u+v)/2$, and this exponent is at least $N(u+v)/2$. Removing the constant term, whose value is exactly 1, we obtain the uniform bound

$$0 \leq B[U_s^2] - 1 \leq P_\infty^{-2} \left\{ (1 - p^{N/2})^{-2} - 1 \right\} \longrightarrow 0. \tag{9.12}$$

Also $B[(U_s-1)^2] \le B[U_s^2]-1$, because $B[U_s] \ge 1$.

In (9.11) we have $r \ge k+1 \ge 2$ and $j \le N-2$. The absolute sum of its scalar coefficients is bounded independently of N, j, r by

$$C_p = \sum_{\ell=0}^{\infty} \frac{p^{2\ell+\binom{\ell}{2}}}{\pi_\ell} < \infty.$$

Let $S_{j,r}$ denote their signed sum, obtained by replacing each $U_{j-\ell}$ by 1. The triangle inequality for the norm $\|f\|_B = B[f^2]^{1/2}$ and (9.12) imply

$$\|R_{j,N} - S_{j,r}\|_B \longrightarrow 0 \quad \text{uniformly for } 0 \le j \le N-2.$$

Consequently $B[R_{j,N}^2]$ is uniformly bounded. For fixed $r \ge 2$, as $N \to \infty$ and $j = N-r$, dominated convergence in the scalar sum and Euler's product identity give

$$S_{j,r} \longrightarrow \sum_{\ell=0}^{\infty} \frac{(-1)^\ell p^{r\ell+\binom{\ell}{2}}}{\pi_\ell} = (p^r;p)_\infty.$$

It follows that

$$B[R_{N-r,N}^2] \longrightarrow (p^r;p)_\infty^2 \qquad \text{for each fixed } r \ge 2. \tag{9.13}$$

An exact simplification of (9.6) yields

$$\frac{A_j^2}{h_j(\alpha_N)} = \frac{(-1)^j p^{-(N^2-3N)/2+r(r-3)/2}}{\pi_j^2 (p^r;p)_j}. \tag{9.14}$$

Substitution into (9.10) shows that

$$\frac{T_{n,k}}{(-1)^{N+1}p^{-(N^2-3N)/2}} = \sum_{r=k+1}^{N} (-1)^r p^{r(r-3)/2} \frac{B[R_{N-r,N}^2]}{\pi_{N-r}^2 (p^r;p)_{N-r}}. \tag{9.15}$$

Each denominator is at least P_∞^3, and the numerators just proved are uniformly bounded. The summable sequence $Cp^{r(r-3)/2}$ therefore dominates the summands. Using (9.13), the right side tends to

$$\begin{aligned} \Sigma_k &= \frac{1}{P_\infty^2} \sum_{r=k+1}^{\infty} (-1)^r p^{r(r-3)/2} (p^r;p)_\infty \qquad (9.16) \\ &= -\frac{p^{-1}}{P_\infty} \sum_{\ell=k}^{\infty} \frac{(-1)^\ell p^{\binom{\ell}{2}}}{\pi_\ell}. \end{aligned}$$

For clarity, the cancellation in this last sum is explicit: Euler's identity at 1 gives the full sum equal to 0, while

$$\sum_{\ell=0}^{k-1} \frac{(-1)^\ell p^{\binom{\ell}{2}}}{\pi_\ell} = \frac{(-1)^{k-1} p^{\binom{k}{2}}}{\pi_{k-1}}.$$

The finite identity follows by subtracting its instances for successive k. Hence

$$\Sigma_k = \frac{(-1)^{k-1} p^{\binom{k}{2}-1}}{P_\infty \pi_{k-1}} \neq 0.$$

Combining this value with (9.15) and $N = n+k$ gives (9.9). □

9.3 Exact center values and their slopes

We use the normalization introduced earlier,

$$F_n(z) = F_{n,0}(z) = \Psi_{n,0}(p^{-n}z), \qquad \Phi_p(z) = \prod_{\ell=1}^{\infty}(1+zp^{\ell})^{\ell}.$$

In particular, theorem 8.3 gives local uniform convergence $F_n \to \Phi_p$, and hence local uniform convergence of every fixed derivative.

Corollary 9.4 (Values and slopes at the limiting zeros)**.** *For every integer $k \geq 1$ and every $n \geq 1$,*

$$F_n(-p^{-k}) = \frac{p^{kn^2}}{\mathcal{A}_{n,0}}. \tag{9.17}$$

For fixed k, as $n \to \infty$,

$$F_n'(-p^{-k}) \sim \frac{(-1)^{k-1}P_\infty}{\mathcal{M}\pi_{k-1}} p^{(k-1)n(n-2)+(-k^2+5k-2)/2}. \tag{9.18}$$

Proof. The Cauchy determinant in the unshifted normalization is

$$\mathcal{C}_{n,0} = p^{n(n+1)(2n+1)/6}\prod_{d=1}^{n-1}(1-p^d)^{2(n-d)}\mathcal{A}_{n,0}.$$

Writing (9.7) in powers of p and cancelling the common factors gives, with $\alpha = p^{-n}z$,

$$\frac{V_n(q,p^{-n}z,0)}{\mathcal{L}_{n,0}(p^{-n}z)\mathcal{C}_{n,0}} = \frac{(-1)^n z^{-n^2}}{\mathcal{A}_{n,0}}. \tag{9.19}$$

The identity is first read away from the vanishing factors and then by cancellation. At $z_k = -p^{-k}$ we have $E_q(p^{-n}z_k) = 0$, and (9.17) follows.

Set $\mu_n(z) = E_q(p^{-n}z)$ and $N = n+k$. Differentiation of its product at the simple zero z_k gives

$$\mu_n'(z_k) = (-1)^{N-1}p^{k-N(N-1)/2}\pi_{N-1}P_\infty. \tag{9.20}$$

The chain rule applied to (9.19) gives the exact relation

$$\frac{F_n'(z_k)}{F_n(z_k)} = \mu_n'(z_k)T_{n,k} - \frac{n^2}{z_k}. \tag{9.21}$$

Now $\mathcal{A}_{n,0} \to \mathcal{M}$ and $\pi_{N-1} \to P_\infty$. Insert (9.9) and (9.20) into (9.21). The term n^2/z_k is negligible compared with $\mu_n'(z_k)T_{n,k}$. Collecting its sign and exponent gives (9.18). □

9.4 Two different displacement scales at the second center

The limiting zero $-p^{-2}$ has multiplicity two in Φ_p. Its two approximating zeros do not have the same displacement scale.

Theorem 9.5 (Asymmetric splitting at the second center)**.** *For all sufficiently large n, F_n has exactly two zeros in a sufficiently small fixed disk about $-p^{-2}$. They are real and simple, and can be labelled so that*

$$z_{n,+} = -p^{-2} + \frac{1-p}{p^2P_\infty}p^{n^2+2n}(1+o(1)), \tag{9.22}$$

$$z_{n,-} = -p^{-2} - \frac{1}{p(1-p)^2P_\infty}p^{n^2-2n}(1+o(1)). \tag{9.23}$$

Equivalently, the corresponding zeros of $H_{n,0}$ are

$$\alpha_{n,+} = -q^{n+2} + \frac{1-p}{P_\infty} p^{n^2+n-2}(1+o(1)), \tag{9.24}$$

$$\alpha_{n,-} = -q^{n+2} - \frac{1}{(1-p)^2 P_\infty} p^{n^2-3n-1}(1+o(1)). \tag{9.25}$$

Proof. Put $z_0 = -p^{-2}$. The product defining Φ_p gives

$$\frac{\Phi_p''(z_0)}{2} = p^4(1-p^{-1}) \prod_{\ell=3}^{\infty} (1-p^{\ell-2})^\ell = -\frac{p^3(1-p)P_\infty^2}{\mathcal{M}}. \tag{9.26}$$

Write

$$a = \mathcal{M}^{-1}, \qquad b = \frac{p^2 P_\infty}{(1-p)\mathcal{M}}, \qquad c = \frac{p^3(1-p)P_\infty^2}{\mathcal{M}};$$

all three constants are positive. The exact center value, the slope asymptotic, and derivative convergence give

$$\begin{aligned} F_n(z_0) &= a p^{2n^2}(1+o(1)), \\ F_n'(z_0) &= -b p^{n^2-2n}(1+o(1)), \\ \frac{F_n''(z_0)}{2} &= -c + o(1). \end{aligned}$$

On a fixed disk about z_0, the third derivatives are uniformly bounded by local uniform convergence and Cauchy's estimates. Thus, uniformly for sufficiently small δ,

$$F_n(z_0+\delta) = F_n(z_0) + F_n'(z_0)\delta + \tfrac{1}{2}F_n''(z_0)\delta^2 + O(\delta^3). \tag{9.27}$$

First let $\varepsilon_n = p^{n^2-2n}$ and set $\delta = \varepsilon_n t$. Division of (9.27) by ε_n^2 gives local uniform convergence in t to

$$-bt - ct^2.$$

Indeed, the constant term after division is $ap^{4n}(1+o(1))$, and the third-order remainder is $O(\varepsilon_n)$ on bounded t-sets. The limiting zero $-b/c$ is simple. Rouché's theorem on a small circle about that zero gives exactly one zero with

$$\delta = -\frac{b}{c}\varepsilon_n(1+o(1)).$$

This is (9.23).

To resolve the zero that is $o(\varepsilon_n)$ from the center, let $\eta_n = p^{n^2+2n}$ and set $\delta = \eta_n u$. Division of (9.27) by p^{2n^2} now gives local uniform convergence in u to

$$a - bu.$$

The quadratic term is $O(p^{4n})$ on bounded u-sets and the third-order term tends to zero. Rouché's theorem about the simple zero a/b gives exactly one zero with

$$\delta = \frac{a}{b}\eta_n(1+o(1)),$$

which is (9.22).

Choose a fixed disk about z_0 containing no other zero of Φ_p. By theorem 8.3 and Rouché's theorem, F_n has precisely two zeros there, counted with multiplicity, for all sufficiently large n. The two zeros just constructed exhaust them and are simple. The coefficients of F_n are real, so conjugation preserves its zeros. Each of the two small disks used above is invariant under conjugation and contains exactly one zero; hence both zeros are real. The factors divided out in the definition of F_n do not vanish near this center. Finally, multiplication by p^{-n} gives (9.24)–(9.25). □

Remark 9.6. The fixed-k slope formula does not by itself determine the full splitting of a limiting zero of multiplicity $k \geq 3$. Such a conclusion would require suitably scaled estimates for higher derivatives. The present section proves the center values and slopes for every fixed k, and the complete two-zero splitting only for $k = 2$.

10 Dimension reduction and the first residual zero for fixed shifts

Throughout this section, $q > 1$, $p = q^{-1}$, and $\pi_j = (p;p)_j$, with $\pi_0 = 1$. We retain the residual functions $\Psi_{n,r}$ and their rescalings

$$F_{n,r}(z) = \Psi_{n,r}(p^{-n-r}z)$$

from theorem 8.3. The shift r is fixed when n tends to infinity. We give an exact reduction of a special value of the original $n \times n$ determinant to an $r \times r$ determinant, and use it to locate the first zero of $\Psi_{n,r}$ beyond the explicitly removed factors.

Lemma 10.1 (A terminating sum and its limit)**.** *For $N \geq 0$ and an integer a, define*

$$S_N(a) = \sum_{\ell=0}^{N} \frac{(-1)^\ell p^{\binom{\ell}{2}+a\ell}\pi_N^2}{\pi_\ell \pi_{N-\ell}^2}. \tag{10.1}$$

For every fixed integer a, including $a \leq 0$,

$$S_N(a) = (p^a;p)_\infty + O_{p,a}(p^N). \tag{10.2}$$

In particular, the limiting value in (10.2) *is zero if $a \leq 0$.*

Proof. For $0 \leq \ell \leq N$, put

$$R_{N,\ell} = \frac{\pi_N^2}{\pi_{N-\ell}^2} = \prod_{h=N-\ell+1}^{N} (1-p^h)^2.$$

Since all factors lie in $[0,1]$,

$$0 \leq 1 - R_{N,\ell} \leq 2 \sum_{h=N-\ell+1}^{N} p^h \leq \frac{2p^{N-\ell+1}}{1-p}. \tag{10.3}$$

Euler's identity, valid for every complex argument, gives

$$(p^a;p)_\infty = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell p^{\binom{\ell}{2}+a\ell}}{\pi_\ell}.$$

The difference contributed by $0 \leq \ell \leq N$ is at most

$$\frac{2p^{N+1}}{1-p} \sum_{\ell=0}^{\infty} \frac{p^{\binom{\ell}{2}+(a-1)\ell}}{\pi_\ell} = O_{p,a}(p^N).$$

This series converges for every fixed a, since $\pi_\ell \geq P_\infty > 0$ and its numerator has quadratic decay in ℓ. The omitted tail of Euler's series is also $O_{p,a}(p^N)$, by the same quadratic decay. This proves the assertion, without a positivity assumption on a. □

We first recall the determinant identity needed for the reduction and include an algebraic proof. For a linear functional ℓ on polynomials, write

$$\Delta_n^{(r)} = \det\big(\ell[x^{i+j+r}]\big)_{0\le i,j<n}.$$

Lemma 10.2 (Confluent Christoffel identity)**.** *Suppose that the moment functional has monic orthogonal polynomials $Q_0, Q_1, \dots, Q_{n+r-1}$ and that the required unshifted moment determinants are nonzero. Then*

$$\frac{\Delta_n^{(r)}}{\Delta_n^{(0)}} = (-1)^{nr} \det\big([x^i]Q_{n+j}(x)\big)_{0\le i,j<r}. \tag{10.4}$$

For $r = 0$, the determinant on the right is interpreted as 1.

Proof. Let $\mathcal{V}_s(u) = \prod_{1\le i<j\le s}(u_j - u_i)$. Apply the tensor product functional $\ell^{\otimes n}$ to the variables $x_1, \dots, x_n$. For distinct auxiliary variables $t_1, \dots, t_r$, the Vandermonde factorization and orthogonality give

$$\frac{\ell^{\otimes n}\big[\mathcal{V}_n(x)^2 \prod_{a=1}^{r}\prod_{i=1}^{n}(t_a - x_i)\big]}{n!\Delta_n^{(0)}} = \frac{\det\big(Q_{n+j}(t_a)\big)_{1\le a\le r,\, 0\le j<r}}{\mathcal{V}_r(t)}. \tag{10.5}$$

Here is a direct justification. Factor

$$\mathcal{V}_{n+r}(x_1, \dots, x_n, t_1, \dots, t_r) = \mathcal{V}_n(x)\mathcal{V}_r(t) \prod_{a=1}^{r}\prod_{i=1}^{n}(t_a - x_i).$$

In the determinant defining the left side, replace the monomials by the monic polynomials $Q_0, \dots, Q_{n+r-1}$. Expand along the n rows containing the x_i. After applying $\ell^{\otimes n}$ against $\mathcal{V}_n(x)$, orthogonality eliminates every selection of columns except $0, \dots, n-1$. The surviving factor is $n!\Delta_n^{(0)}$, which proves (10.5).

Let all t_a tend to zero. The quotient on the right tends to the coefficient determinant in (10.4); this follows by expanding each $Q_{n+j}(t_a)$ at zero and taking the lowest alternating term. On the left, the inserted product becomes $(-1)^{nr}\prod_i x_i^r$. Expanding the two Vandermonde determinants shows that its image under $\ell^{\otimes n}$ is $(-1)^{nr}n!\Delta_n^{(r)}$. The result follows. □

Theorem 10.3 (Exact reduction at the first limiting zero)**.** *For integers $n \ge 1$ and $r \ge 0$, put*

$$R_r = \frac{r(r-1)}{2}, \qquad T_r = \frac{r(r-1)(r-2)}{6},$$

and define the $r \times r$ matrix

$$B_{n,r} = \big(p^{ij} S_{n+j}(r+1-i-j)\big)_{0\le i,j<r}. \tag{10.6}$$

Then

$$\boxed{\mathcal{A}_{n,r}F_{n,r}(-q) = \frac{(-1)^{R_r} p^{n^2 - T_r}}{\prod_{i=0}^{r-1}\pi_i} \det B_{n,r}.} \tag{10.7}$$

Equivalently, the residual function in (10.7) *is evaluated at $\alpha = -q^{n+r+1}$. Empty determinants and products give the exact identity*

$$\mathcal{A}_{n,0}F_{n,0}(-q) = p^{n^2}. \tag{10.8}$$

Proof. Write $W_n^{(r)}(\alpha) = V_n^{(r)}(q, \alpha, 0)$ and set

$$m = n + r + 1, \qquad \alpha = -p^{-m}.$$

At this value of α, the product formula for E_q gives $E_q(\alpha) = 0$, so the actual determinant agrees with $W_n^{(r)}(\alpha)$. The monic polynomials for the moment functional with moments $-v_k(q, \alpha, 0)$ are denoted by P_N^α as in Lemma 9.1. Multiplication of all moments by -1 has no effect on the quotient of two determinants of the same size. Thus Lemma 10.2 gives

$$\frac{W_n^{(r)}(\alpha)}{W_n^{(0)}(\alpha)} = (-1)^{nr} \det\bigl([x^i] P_{n+j}^\alpha(x)\bigr)_{0 \le i,j < r}. \tag{10.9}$$

The identity follows first for positive α, where the moment functional is positive, and then as an identity of rational functions of α. At the present specialization the needed unshifted determinants are nonzero: their factors $1 + \alpha p^h$ have $h < m$, by Proposition 9.2.

For $N = n + j$, the terminating formula in Lemma 9.1, with the summation index reversed, becomes

$$\begin{aligned} P_N^\alpha(x) &= a_N \sum_{\ell=0}^{N} \frac{(-1)^\ell p^{\binom{\ell}{2} + (m-N)\ell} \pi_N^2}{\pi_\ell \pi_{N-\ell}^2} (-xp^m; p)_{N-\ell}, \\ a_N &= p^{-N(N-1)/2 - mN}. \end{aligned} \tag{10.10}$$

Divide all the polynomials in the coefficient determinant by the common power series $G(x) = (-xp^m; p)_\infty$, whose constant term is 1. For the first r coefficients this division is multiplication by a lower triangular matrix with diagonal entries 1; hence it leaves their determinant unchanged. Euler's reciprocal-product expansion gives

$$\frac{(-xp^m; p)_{N-\ell}}{(-xp^m; p)_\infty} = \sum_{i=0}^{\infty} \frac{(-1)^i x^i p^{i(m+N-\ell)}}{\pi_i}.$$

Consequently,

$$[x^i] \frac{P_N^\alpha(x)}{G(x)} = a_N \frac{(-1)^i p^{i(m+N)}}{\pi_i} S_N(m - N - i). \tag{10.11}$$

Extracting the row and column factors in (10.11) yields

$$\det\bigl([x^i] P_{n+j}^\alpha(x)\bigr)_{i,j<r} = \frac{(-1)^{R_r} p^{-rn(n-1)/2 - mrn - T_r}}{\prod_{i=0}^{r-1} \pi_i} \det B_{n,r}. \tag{10.12}$$

For example, the exponent here is obtained from

$$\sum_{j=0}^{r-1} \left(-\frac{(n+j)(n+j-1)}{2} - m(n+j) \right) + (m+n) \sum_{i=0}^{r-1} i = -\frac{rn(n-1)}{2} - mrn - T_r.$$

For completeness, we track the remaining normalization. Put

$$d_n = \frac{n(n-1)(4n+1)}{6}, \quad J_n = \prod_{h=1}^{n-1} (1 - p^h)^{2(n-h)}, \quad L_n(\alpha) = \prod_{h=1}^{n-1} (1 + \alpha p^h)^{n-h}.$$

The zero-value product and the Cauchy determinant evaluation read

$$W_n^{(0)}(\alpha) = (-1)^n p^{-d_n} J_n L_n(\alpha), \qquad \mathcal{C}_{n,r} = p^{K_{n,r}} J_n \mathcal{A}_{n,r}.$$

Since $d_n + K_{n,r} = n^3 + rn(n+1)/2$ and $\mathcal{L}_{n,r}(\alpha) = \alpha^{n(n+r)} L_n(\alpha)$, substitution of (10.9) into the definition of $\Psi_{n,r}$ gives

$$\mathcal{A}_{n,r} F_{n,r}(-q) = p^{-n^3 - rn(n+1)/2 + mn(n+r)} \det\big([x^i] P^{\alpha}_{n+j}(x)\big)_{i,j<r}. \tag{10.13}$$

The signs cancel because $n + n(n+r) + nr = n(n+1) + 2nr$ is even. Combining (10.12) and (10.13), the remaining exponent is

$$-n^3 - \frac{rn(n+1)}{2} + mn(n+r) - \frac{rn(n-1)}{2} - mrn - T_r = n^2 - T_r,$$

where $m = n + r + 1$. This proves (10.7). □

Corollary 10.4 (Asymptotic value at the center)**.** *For every fixed integer $r \geq 0$, let*

$$C_r = \prod_{a=2}^{r+1} (p^a; p)_\infty = \frac{P_\infty^r}{\prod_{j=1}^r \pi_j}, \qquad C_0 = 1. \tag{10.14}$$

Then

$$\mathcal{A}_{n,r} F_{n,r}(-q) = p^{n^2} C_r \big(1 + O_{p,r}(p^n)\big). \tag{10.15}$$

Proof. By Lemma 10.1, the entries of $B_{n,r}$ converge at rate $O_{p,r}(p^n)$ to those of the matrix

$$B_{\infty,r} = \big(p^{ij} (p^{r+1-i-j}; p)_\infty\big)_{0 \leq i,j < r}.$$

For every entry, including those with a vanishing infinite product,

$$p^{ij} (p^{r+1-i-j}; p)_\infty = (p^{r+1-j}; p)_\infty \prod_{h=1}^{i} (p^j - p^{r+1-h}). \tag{10.16}$$

After the first factor is extracted from each column, row i consists of the values at $1, p, \ldots, p^{r-1}$ of a monic polynomial of degree i. Its determinant is therefore the Vandermonde determinant. Thus

$$\begin{aligned} \det B_{\infty,r} &= C_r \prod_{0 \leq j < k < r} (p^k - p^j) \\ &= (-1)^{R_r} p^{T_r} C_r \prod_{i=0}^{r-1} \pi_i. \end{aligned} \tag{10.17}$$

In particular this determinant is nonzero. Since r is fixed, (10.1) implies $\det B_{n,r} = \det B_{\infty,r}(1 + O_{p,r}(p^n))$. The conclusion now follows from theorem 10.3. □

Theorem 10.5 (First residual zero for each fixed shift)**.** *Fix $q > 1$ and an integer $r \geq 0$. For all sufficiently large n, $\Psi_{n,r}$ has a unique zero of smallest modulus. It is simple, real, and negative. Denote it by $\beta_{n,r}$. Then*

$$\boxed{\beta_{n,r} = -q^{n+r+1} - \frac{q^{-n^2+n+r+1}}{P_\infty \pi_r} \big(1 + O_{q,r}(q^{-n})\big).} \tag{10.18}$$

In particular, $\beta_{n,r} < -q^{n+r+1}$ for all sufficiently large n. The estimate concerns each fixed r; its implicit constant is allowed to depend on r.

Proof. The limiting function in theorem 8.3 is

$$\Phi_p(z) = \prod_{\ell=1}^{\infty} (1 + zp^{\ell})^{\ell}.$$

Its first zero is the simple zero $-q$, and the next zero is $-q^2$. Choose c with $q < c < q^2$. Uniform convergence on $|z| = c$ and Rouché's theorem show that $F_{n,r}$ has exactly one zero, counted with multiplicity, in $|z| < c$ for all sufficiently large n. This zero, denoted $z_{n,r}$, tends to $-q$. It is simple, and it is real because $F_{n,r}$ takes real values on the real axis and its nonreal zeros occur in conjugate pairs. It is consequently the unique zero of smallest modulus.

The products defining $\mathcal{A}_{n,r}$ and $\mathcal{A}_{\infty,r}$ give, for fixed r,

$$\mathcal{A}_{n,r} = \mathcal{A}_{\infty,r}\big(1 + O_{p,r}(p^n)\big). \tag{10.19}$$

Indeed, the sum of $p^{r+i+j+1}$ over the omitted pairs with $i \geq n$ or $j \geq n$ is $O_{p,r}(p^n)$, which also bounds the logarithmic product tail. Therefore Corollary 10.4 gives

$$F_{n,r}(-q) = \frac{C_r}{\mathcal{A}_{\infty,r}} p^{n^2}\big(1 + O_{p,r}(p^n)\big) > 0 \tag{10.20}$$

for all sufficiently large n.

Differentiating the limiting product at its simple zero yields

$$\Phi_p'(-q) = \frac{pP_\infty}{\mathcal{M}} > 0. \tag{10.21}$$

By the derivative form of theorem 8.3, $F_{n,r}'$ differs from Φ_p' by $O_{p,r}(p^n)$ on a fixed neighborhood of $-q$. The real mean value theorem first gives $z_{n,r} + q = O_{p,r}(p^{n^2})$, since the derivative there is bounded away from zero. Applied once more with the derivative estimate, it gives

$$\begin{aligned} z_{n,r} + q &= -\frac{\mathcal{M}C_r}{P_\infty \mathcal{A}_{\infty,r}} p^{n^2-1}\big(1 + O_{p,r}(p^n)\big) \\ &= -\frac{p^{n^2-1}}{P_\infty \pi_r}\big(1 + O_{p,r}(p^n)\big). \end{aligned} \tag{10.22}$$

For the second equality, cancellation of the convergent products gives

$$\frac{\mathcal{M}C_r}{\mathcal{A}_{\infty,r}} = \frac{1}{\pi_r}. \tag{10.23}$$

Finally $\beta_{n,r} = p^{-n-r} z_{n,r}$, which proves (10.18), including the sign of the displacement. □

11 Conclusions and remaining questions

The formal moment representation determines the generic polynomial order, and the positive geometric representation determines the actual cubic decay. With the prescribed cyclotomic divisor, these yield an explicit restriction on proportional shifts: the arithmetic comparison cannot improve its unshifted critical values 12/7 and 6.

The analytic normalization retains information beyond these cubic rates. Its Schur expansion gives a uniform entire scaling limit and a fixed-parameter asymptotic with its first correction. Classical q-Meixner orthogonality then evaluates determinant specializations that are much smaller than the uniform error in that limit. Combining these two scales gives the first zero displacement for every fixed shift and the unequal displacements in the second

unshifted cluster. The finite-dimensional reduction is also an exact identity: for fixed r, the size of the remaining determinant is independent of n.

Several questions remain outside these results. The exact generic shifted cyclotomic valuations are not determined by the guaranteed divisor. Additional arithmetic content, different determinants, or sharper conjugate estimates could change the norm comparison. For the analytic problem, the individual splitting of clusters of multiplicity at least three requires further information beyond the center value and first derivative. Uniform zero displacement estimates when r grows with n likewise do not follow from the fixed-r determinant asymptotic.

The modular specializations in appendix A are applications of existing transcendence theorems. They are independent of the shift barrier and the refined zero analysis, and are not claimed as newly resolved irrationality problems.

A Modular specializations: consequences of existing theory

This appendix is logically independent of the determinant estimates. It records applications of classical modular identities and transcendence results, not a new proof of the underlying modular transcendence theorem.

For $p = e^{2\pi i\tau}$, $\operatorname{Im}\tau > 0$, let $J(p) = j(\tau)$. The theorem of Barré-Sirieix, Diaz, Gramain, and Philibert implies that $J(p)$ is transcendental for algebraic $0 < |p| < 1$. The same conclusion follows from the stronger result in [5, Corollary 2]; that paper also identifies the earlier 1995 theorem in its historical discussion. Weber's product and algebraic identity, in the normalization of [6, (9.9)–(9.11)], are

$$w(\tau) = \sqrt{2}\,p^{1/24}\prod_{n\geq 1}(1+p^n), \qquad (w^{24}+16)^3 = j(\tau)w^{24}. \tag{A.1}$$

Only the twenty-fourth power is used, so the choice of root has no effect.

Proposition A.1. *If $q \in \overline{\mathbb{Q}}$ and $|q| > 1$, then $E_q(1)$ is transcendental. If $u \in \overline{\mathbb{Q}}$, $|u| > 1$, and $m \in \mathbb{Z}$, then*

$$E_{u^2}(\pm u^{2m+1}) \quad \textit{is transcendental.} \tag{A.2}$$

Proof. Put $p = q^{-1}$, $\mu = E_q(1)$, and $z = p\mu^{24}$. The nonvanishing product (4.2) and (A.1) give

$$J(p) = \frac{(1+256z)^3}{z}. \tag{A.3}$$

If μ were algebraic, then so would be $z \neq 0$ and $J(p)$, contrary to the cited theorem.

To extend the family, splitting absolutely convergent products into even and odd indices yields

$$E_{u^2}(u)E_{-u}(1) = 1, \qquad E_{u^2}(-u)E_u(1) = 1. \tag{A.4}$$

For example, writing $x = u^{-2}$, the first product is

$$\prod_{n\geq 1}(1-x^{2n-1})\prod_{n\geq 1}(1+x^n) = 1,$$

because $\prod(1+x^n) = \prod(1-x^{2n})/\prod(1-x^n)$. The second identity follows by the same separation with u^{-1}. Both reciprocal values are therefore transcendental. Iterating (4.1) in both directions multiplies these values by nonzero algebraic factors to obtain (A.2). Indeed, each factor is $1 \pm u^{2j+1}$ for some integer j, which is nonzero because $|u| > 1$. □

For rational $u = a/b > 1$ in lowest terms, $q = u^2$ and $\gamma_q = \log a/\log b$. Choosing $a = b+1$ makes this ratio approach one. These are restricted pairs (q, α), not arbitrary independent rational parameters. In particular, neither this appendix nor the shift theorem asserts a new uniform critical value below $12/7$.

B Exact checks and reproducibility

The proofs above are symbolic and analytic; finite computations are used only to check signs, indices, and specializations. For the revised manuscript, the following polynomial orders were checked by exact fraction-free elimination after generating the entries from (1.2).

Table 1: Exact orders at $q = 0$ for $\alpha = 2$ and $\mu = 5$.

n	r	Formula (1.5)	Computed order
3	4	5	5
4	3	10	10
5	2	18	18
5	5	30	30
6	4	44	44

At $\alpha = 2$, $\mu = 3$, $n = 5$, $r = 2$, the order is 19 instead of the generic value 18, as predicted by the vanishing lowest coefficient in Remark 2.3. The exact $\mu = 0$ product was checked at $q = 2$, $\alpha = 2$, $1 \le n \le 6$. The shifted dimension-reduction identity was checked against direct rational Hankel determinants at $q = 2$, $1 \le n \le 4$, $0 \le r \le 4$.

These checks can be reproduced without additional data: form $v_0 = \mu - 1$, apply the recurrence through index $2n - 2 + r$, and compute the indicated determinant over $\mathbb{Q}[q]$ or $\mathbb{Q}$. To check the reduction identity, set $\alpha = -2^{n+r+1}$ and $\mu = 0$, compute its finite sums $S_N(a)$ in rational arithmetic, and compare both sides after the stated normalization. No numerical experiment is used to establish a transcendence assertion, the location of an individual asymptotic zero, or the exactness of an unspecified cyclotomic divisor. The single source file contains the entire manuscript and bibliography and requires no external data files.

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